Separable Differential Equations with Initial Conditions

Separating and integrating, applying the initial condition to pin down the constant, and using the particular solution

Directions. Give the particular solution — an equation with no arbitrary constant left in it — unless a problem asks for something else.

- **Separate:** get every y (with dy) on one side and every x (with dx) on the other.
- **Integrate both sides** and put a single $+C$ on one side only.
- **Apply the initial condition immediately**, before solving for y if that is easier — it usually is.
- Round decimal answers to two places and keep the units.

1. Solve $\dfrac{dy}{dx} = 6x^2$ with $y(0) = 1$. Give the particular solution, then use it to find $y(2)$.

Answer: ________________

2. Solve $\dfrac{dy}{dx} = 3x^2 - 4$ with $y(2) = 5$. Find the value of C explicitly before writing the particular solution.

Answer: ____________________

3. Solve $\dfrac{dy}{dx} = xy$ with $y(0) = 2$. Separate first, integrate both sides, then exponentiate to isolate y. Use your particular solution to find $y(2)$, rounded to two decimal places.

Answer: ____________________

4. Solve $\dfrac{dy}{dx} = \dfrac{2x+1}{y}$ with $y(0) = 3$. Find C from the initial condition, then give y as an explicit function of x and compute $y(2)$ to two decimal places.

Answer: ________________

5. Solve $\dfrac{dy}{dx} = y^2$ with $y(1) = 1$. Give the particular solution and evaluate it at $x = 1.5$. For which values of x is your solution actually defined?

Answer: ________________

6. A 80 milligram sample of a radioactive isotope decays at a rate proportional to the amount present, so $\frac{dA}{dt} = -kA$, and its half-life is 10 days. Find k from the half-life, then find how much of the sample remains after 25 days, in milligrams, to two decimal places.

Answer: ________________ mg

7. Solve $\frac{dy}{dx} = y \cos x$ with $y(0) = 4$. Give the particular solution, then evaluate it at $x = 1$ radian, to two decimal places.

Answer: ________________

8. Solve $\dfrac{dy}{dx} = \dfrac{3x^2}{2y}$ with $y(1) = 2$. Separate, integrate, apply the condition, and give y explicitly. Then find $y(3)$ to two decimal places, taking the branch the initial condition selects.

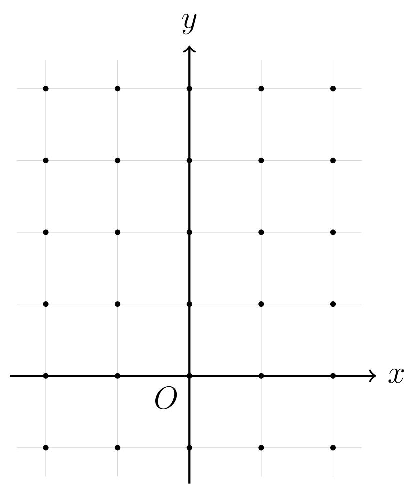
Answer: ___________________

9. A population grows so that $\dfrac{dP}{dt} = 0.08P$, with $P(0) = 500$ and t measured in years. Solve the equation, then find how long it takes the population to reach 1500, to two decimal places.

Answer: ___________________ years

10. Consider the logistic equation $\dfrac{dy}{dx} = y(3-y)$.

(a) Find every equilibrium solution by setting $\frac{dy}{dx} = 0$. **(b)** On the lattice below, draw a short segment of the correct slope at each dot, mark the equilibrium solutions as horizontal lines, and write one sentence describing what happens to the solution through the point one unit above the origin as x grows. Each square is one unit and the origin is marked O.



Answer: ____________________